

Current Features:

1. Authentication & Tenancy

- Email/password auth (Better Auth), optional signup approval + platform admin (Sec 4.2 #1)
- Multi tenant organizations, memberships, role scoped API access (Sec 4.2 #2, Sec 5.3)
- Every product row scoped by org_id ..100% cross tenant isolation (NFR 02)

2. Agent Management

- Agent CRUD, versioning, deploy (Sec 4.2 #3)
- Multiple voices and languages support, including multilingual (EN/UR/auto)
- Sandbox microphone testing.. consent based, no auto start (Sec 4.2 #4)

3. Knowledge Base (RAG)

- Text, PDF, DOCX ingestion (Sec 4.2 #5, Sec 6.2)
- Chunking (~800 chars), local embeddings, optional Qdrant mirror
- Search modes: qdrant | hybrid | keyword
- Original file discarded after extraction (privacy conscious design)

4. Tools Studio

- HTTP tools: configurable method, URL, headers, body templates (Sec 4.2 #6, Sec 6.3)
- Test invocations directly from the dashboard

5. Automation Packs

- 3 pre built vertical packs: **Restaurant, Receptionist, Appointments** (Sec 4.2 #7)
- Postgres backed executors, type=pack tools
- Webhooks: signed delivery, worker retries

6. Workflow Canvas

- n8n inspired drag and drop conversation graph builder (Sec 4.2 #8, Sec 6.4)
- Workspace persisted, connect nodes/ports
- Note: canvas is UI only; execution engine is still on the roadmap

7. Guardrails

- PII caution, topic bounds, jailbreak refusal, language allow list (Sec 4.2 #9, Sec 6.5)
- Persisted to agents.guardrails JSONB and injected into the live turn prompt

8. Conversations & Analytics

- Conversation history, transcripts (Sec 4.2 #10)
- Analytics rollups (SQL based, in dashboard)

9. Credits & Billing

- Credit wallet system (creditBalanceCents) (Sec 4.2 #11, Sec 6.6)
- Immutable ledger (credit_ledger)
- Voice turns metered per debit; STT/LLM/TTS/tools tracked separately
- Platform admin credit grants; Stripe schema ready (live checkout not yet complete)

10. Developer Surfaces

- Org API keys (vfk_ prefix) ..server to server calls (Sec 4.2 #12, Sec 6.7)
- Embed tokens (vw_ prefix) ..public widget, allowlisted origins
- Public docs + OpenAPI 3.1 spec (Sec 4.2 #13)

11. Privacy & Compliance Primitives

- Privacy data export endpoint: /api/orgs/:orgId/privacy/export (Sec 4.2 #13, Appendix C)
- Documented threat model
- Health reporting for commercial + open source backends (Sec 4.2 #14)

12. Voice Pipeline (Core Engine)

- 9 step turn pipeline: credit gate → load agent → persist message → retrieve knowledge → inject guardrails → run LLM → execute tools → synthesize speech → stream + debit (Sec 6.1)
- Default stack: ElevenLabs Scribe (STT) + Groq Llama 3.3 (LLM) + ElevenLabs Flash (TTS), Gemini fallback (Sec 9.1)
- Optional self hosted path: Coqui XTTS (TTS), Qdrant (vectors) (Sec 9.2)

13. Admin & Operations

- Platform admin: approve tenants, grant credits, inspect usage
- Service health endpoints with provider flags (Sec 13)
- Structured API error logs, worker logs

14. Security

- Server only provider secrets (never VITE_*) (NFR 03)
- Tool egress URL allowlist + SSRF guards (NFR 04)
- Redis backed rate limiting (NFR 05)

- Compose healthchecks, TLS via Caddy (NFR 06)

Explicitly NOT Included in v1 (Limitations, Sec 16)

- Telephony (PSTN/SIP) ..deferred (ADR 004)
- SOC 2 / HIPAA certification ..roadmap only
- Full Stripe live checkout ..schema ready, not complete
- Workflow graph execution engine ..canvas UI only
- SSO/SAML ..future work
- Multi AZ high availability ..single EC2 host only

FEATURES TO ADD...

1. One click integrations for common non tech tools

Instead of "Connect Custom API," add pre wired connectors for tools small businesses already use:

- Google Sheets (log leads/bookings)
- WhatsApp Business
- Google Calendar
- Email (Gmail/Outlook notification on new booking)

This is a much bigger unlock for target market (South Asian SMBs) than raw HTTP tools

2. Human readable analytics (not just latency/ms numbers)

Analytics currently shows technical metrics (latency, conversation counts). Add a plain business view:

- "You saved ~X hours this week"
- "12 bookings captured while you were busy"
- "Missed call recovery rate: 90%"

ISSUES:

1. Pre built Agents/Personas not showing on Agents page

- Dashboard/Agents
- **Report says:** 3 Automation Packs are explicitly ready ..Restaurant (reservations), Receptionist (front desk/call routing), Appointments (booking/intake) (Sec 4.2 #7, Sec 6.3).

- **Issue:** Only manually created shows up on the Agents page. The 3 report documented pre built personas are not visible or selectable there.. even though the Workflows page shows them as installable packs...after installing a pack, testing the resulting agent in Sandbox doesn't return correct/expected responses the pack logic doesn't seem to be wired into the live turn pipeline properly

2. No voice preview option

- VoiceLibrary
- **Issue:** There's no way to listen to a voice sample before assigning it to an agent.

3. Voice selection not actually applying

No matter which voice is selected during agent setup, the agent always speaks in the same default voice... not the one that was chosen.

4. Voice/audio drops after greeting ..conversation continues as text only

- Sandbox (live call)
- **Issue:** Agent speaks the greeting correctly, but after the user responds the reply comes back as text in the transcript only.. no audio plays for that response or afterward.

5. Knowledge base documents not linked to a specific agent

- Knowledge Base page
- **Issue:** Files/documents can be uploaded successfully, but there's no option to assign/allocate a specific document to a specific agent

6. Tools not configuring/saving properly

- Tools & Integrations page
- **Issue:** Attempting to configure a custom API tool (or install one from the integrations library) doesn't save or take effect correctly

9. Fallback LLM not configured

- **Where:** Integrations page → "Fallback LLM: Not configured"
- **Report says:** Section 9.1 lists Gemini 2.5 Flash as the automatic failover LLM in the default production path.

10. Sandbox running on fallback path, not primary stack

- **Where:** Sandbox → shows STT: BROWSER and LLM: FALLBACK

- **Report says:** Default live path should use ElevenLabs Scribe (STT) and Groq Llama 3.3 (LLM).
- **Issue:** Live testing is currently running on browser native speech recognition and a fallback LLM, not the primary stack described in the report. This means the sub 500ms latency claim (NFR 01) has never actually been tested on the real pipeline.
- **Issue:** GEMINI_API_KEY is missing from the environment, so the failover path is non functional.

11. Open source backends (Coqui/Qdrant) have no UI visibility

- **Where:** Not present anywhere in tenant dashboard.
- **Report says:** Section 5.4 ..this is intentional; these are platform level, server side config (TTS_PROVIDER, QDRANT_URL), not tenant facing.
- **Issue:** Not a bug, but there's no way to demonstrate this capability visually during a defense/demo ..it only exists in .env.

12. Privacy export endpoint has no dashboard entry point

- **Where:** Settings page ..no "Export my data" button visible.
- **Report says:** Section 4.2 (#13) and Appendix C confirm GET /api/orgs/:orgId/privacy/export exists as a functional requirement.
- **Issue:** The endpoint may exist on the backend, but there's no discoverable UI trigger for it.